

Reconstruction error from a digitized Kaplan-Meier curve was too small to change which method ranks best for RMST

A simulation study answering catalog problem DIA-10

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Abstract

Background. Simulations of population-adjusted survival comparisons use true individual data for the comparator, while real analyses reconstruct it from a published Kaplan-Meier figure. If the omitted reconstruction error differs between methods it changes their ranking. Catalog problem DIA-10 asks whether it does.

Methods. A 250-patient comparator arm published as a pixel-rounded Kaplan-Meier figure with a 6-monthly risk table or none, spread or clustered censoring, at fine or coarse resolution, and reconstructed by Guyot's algorithm (1). Kaplan-Meier and Weibull RMST to 24 months on true and reconstructed data, and the digitized curve integrated directly; 1000 replicates per cell with common random numbers.

Results. The better of the two methods was the same with and without reconstruction in all 8 cells (Weibull, correctly specified here). Reconstruction changed RMSE by at most 1%: it added a bias of 0.003 to 0.040 months to Kaplan-Meier RMST and 0.006 to 0.059 to Weibull RMST, against a sampling SD of about 0.5 months. Integrating the digitized curve directly did as well as reconstructing.

Conclusion. For RMST within follow-up, omitting reconstruction error from a simulation does not misattribute error to methods. The larger exposure of the likelihood-based method was visible but immaterial.

1 The problem

The reported error of each method in a simulation that omits an access mechanism is its error without that mechanism. Rankings survive only if the omitted error is common to the methods. Reconstruction should expose a likelihood fitted to individual event and censoring times more than a Kaplan-Meier summary, because the reconstruction preserves the curve but not where censoring falls.

2 Design

Registered protocol: `protocol.md`. The data-generating and publication model is OUT-13's: Weibull event times (shape 1.2, median 18 months), 24 months of accrual and 36 of follow-up, 30% of patients also dropping out before 6-monthly visits in the clustered pattern; the curve read at pixel-column centers of a 1500 by 1000 or 300 by 200 plot and rounded to the pixel grid. The individual-data arm of an unanchored RMST difference is common to all methods and omitted.

3 Results

Table 1: RMSE of the comparator arm’s RMST to 24 months against the population value (months); 1000 replicates per cell.

resolu- tion	risk table	censoring	KM, true data	KM, recon- structed	Weibull, true data	Weibull, re- constructed	curve integrated
fine	6- monthly	spread	0.518	0.517	0.506	0.506	0.517
coarse	6- monthly	spread	0.507	0.505	0.501	0.501	0.506
fine	none	spread	0.520	0.520	0.506	0.506	0.520
coarse	none	spread	0.513	0.512	0.501	0.502	0.511
fine	6- monthly	clustered	0.531	0.530	0.518	0.517	0.530
coarse	6- monthly	clustered	0.532	0.531	0.518	0.520	0.530
fine	none	clustered	0.518	0.518	0.500	0.499	0.518
coarse	none	clustered	0.513	0.513	0.504	0.506	0.513

The registered primary was refuted (Table 1): no ranking changed and no RMSE moved by 5%. The reconstruction’s bias was upward and largest at coarse resolution, because a drop read at a pixel column’s center is recorded up to a column late; the Weibull fit carried more of it than the Kaplan-Meier summary, as predicted, but the difference was a few hundredths of a month.

4 What this does not answer

Reconstruction error only; RMST within follow-up only, where OUT-13’s probes found reconstruction error smallest; extrapolated survival and hazard ratios, where censoring placement matters more, were not scored for ranking; no human digitization variability. Peer review has not been done.

References

1. Patricia Guyot, A. E. Ades, Mario J. N. M. Ouwens, Nicky J. Welton. Enhanced secondary analysis of survival data: Reconstructing the data from published Kaplan-Meier survival curves. BMC Medical Research Methodology. 2012;12:9. doi:[10.1186/1471-2288-12-9](https://doi.org/10.1186/1471-2288-12-9)